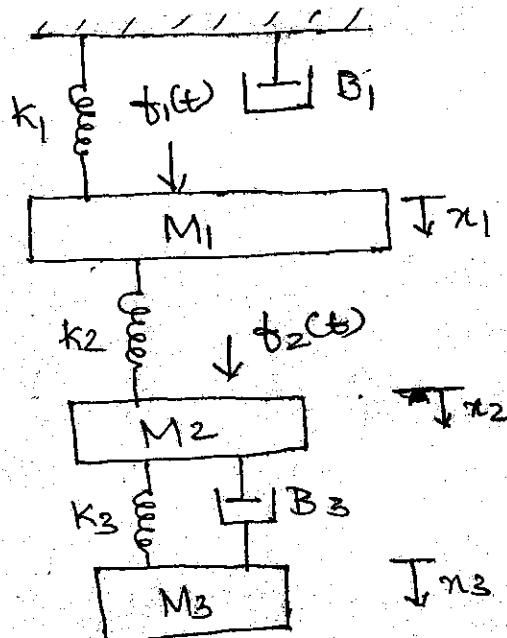


B. Tech Degree VI Semester Examination, April 2010**ME 603 INSTRUMENTATION THEORY AND CONTROL ENGINEERING**
(2002 Scheme)

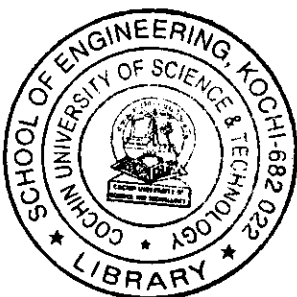
Time : 3 Hours

Maximum Marks : 100

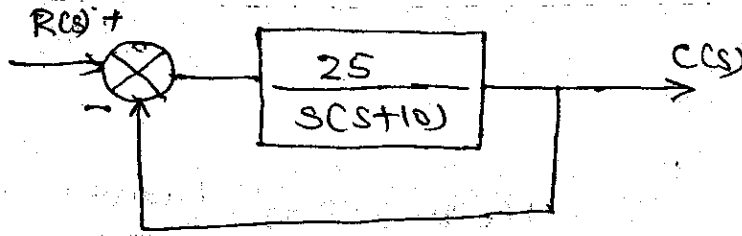
- I. (a) Develop a mathematical model for a second order system. Discuss on its step response with the help of relevant plots and equations. (8)
- (b) Discuss the various types of errors. A 600V voltmeter is specified to be accurate within $\pm 2\%$ off full scale. Calculate the limiting error when the instrument is used to measure 250V. (7)
- (c) Describe the following terms.
- | | |
|--------------------|-------------------------|
| (i) loading effect | (ii) backlash |
| (iii) dead space | (iv) static sensitivity |
| (v) threshold | |
- (5)
- OR**
- II. (a) With the help of a neat block diagram, explain the functional elements of a measurement system with pressure thermometer as an example. (7)
- (b) Explain seismic instrument as a second order instrument. (6)
- (c) A liquid thermometer has a glass bulb protected by a well. The system can be represented by a double capacity system with time constants of 40S for the well and 20S for the bulb. The thermometer is subjected to a cyclic change of $\pm 10^\circ\text{C}$ which occurs every 120S. Find the maximum value of the indicated temperature and delay time. (7)
- III. (a) What are strain rosettes? What do you mean by temperature compensation in strain gauge. (7)
- (b) With the help of neat sketches explain the working of hydraulic and pneumatic load cells. (6)
- (c) Explain –
- | | |
|--|-----|
| (i) Heenan and Froude hydraulic dynamometer | (7) |
| (ii) Beam and strain gauge transmission dynamometer. | (7) |
- OR**
- IV. (a) Write notes on optical pyrometers and total radiation pyrometers. (7)
- (b) Explain
- | | |
|-------------------------|-----|
| (i) Infrared pyrometry | (7) |
| (ii) Gas Chromatography | (7) |
- (c) Explain the working principle of
- | | |
|-----------------------------|-----|
| (i) Sound level meter | (6) |
| (ii) Geiger Muller Counter. | (6) |
- V. (a) Write the differential equations governing the mechanical system shown in the figure. Draw force voltage and force current electrical analogous circuits. (7)



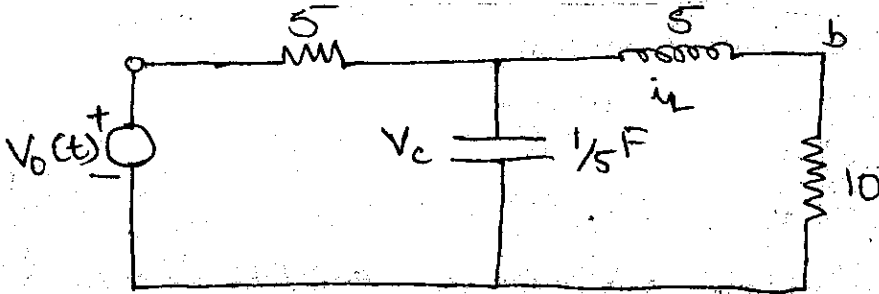
(Turn Over)



- (b) Derive the transfer function of the closed loop system given below. (6)



- (c) Write the state equation in matrix form for the circuit given below. (7)



OR

- VI. (a) The closed loop transfer function of a system is given by $\frac{C(s)}{R(s)} = \frac{7}{s^2 + 4s + 5}$. Determine. (i) the damping ratio (ii) natural frequency (iii) step response (iv) error response (6)
- (b) Evaluate the dynamic error co-efficients for the unity feed back system with forward path transfer function. $G(s) = \frac{15}{s(s+2)}$. (7)
- (c) The open loop transfer function of a unity feed back control system is given by $G(s) = \frac{0.4s+1}{s(s+0.6)}$. Determine the transient response for unit step input and sketch the response. Evaluate the maximum overshoot and the corresponding peak time. (7)
- VII. (a) What is Laiponov function? How it is being developed. Explain how it can be used in the stability analysis of the system. (10)
- (b) Using Routh criterion investigate the stability of a unity feed back control system whose open loop transfer function is given by $G(s) = \frac{e^{-sT}}{s(s+2)}$. (10)

OR

- VIII. (a) Draw the Bode plot for the transfer function $G(s) = \frac{50}{s(1+0.25s)(1+0.1s)}$. From the graph determine (i) Gain cross over frequency (ii) Phase cross over frequency (iii) G.M. and P.M (iv) Stability of the system. (12)
- (b) Explain Absolute and relative stability. (8)
- IX. (a) Describe the transfer function of AC servo motor and armature controlled DC servomotor and explain. (7)
- (b) Explain the operating principle, application, types and characteristics of stepper motor. (7)
- (c) Explain the construction, working principle and applications of synchros. (6)

OR

- X. (a) Write notes on : (i) Pneumatic controllers (ii) Hydraulic controller (iii) Integral control action (iv) Derivative control action (8)
- (b) Write notes on tachogenerators. (5)
- (c) What are the advantages and limitations of derivative control action compared to others. (7)